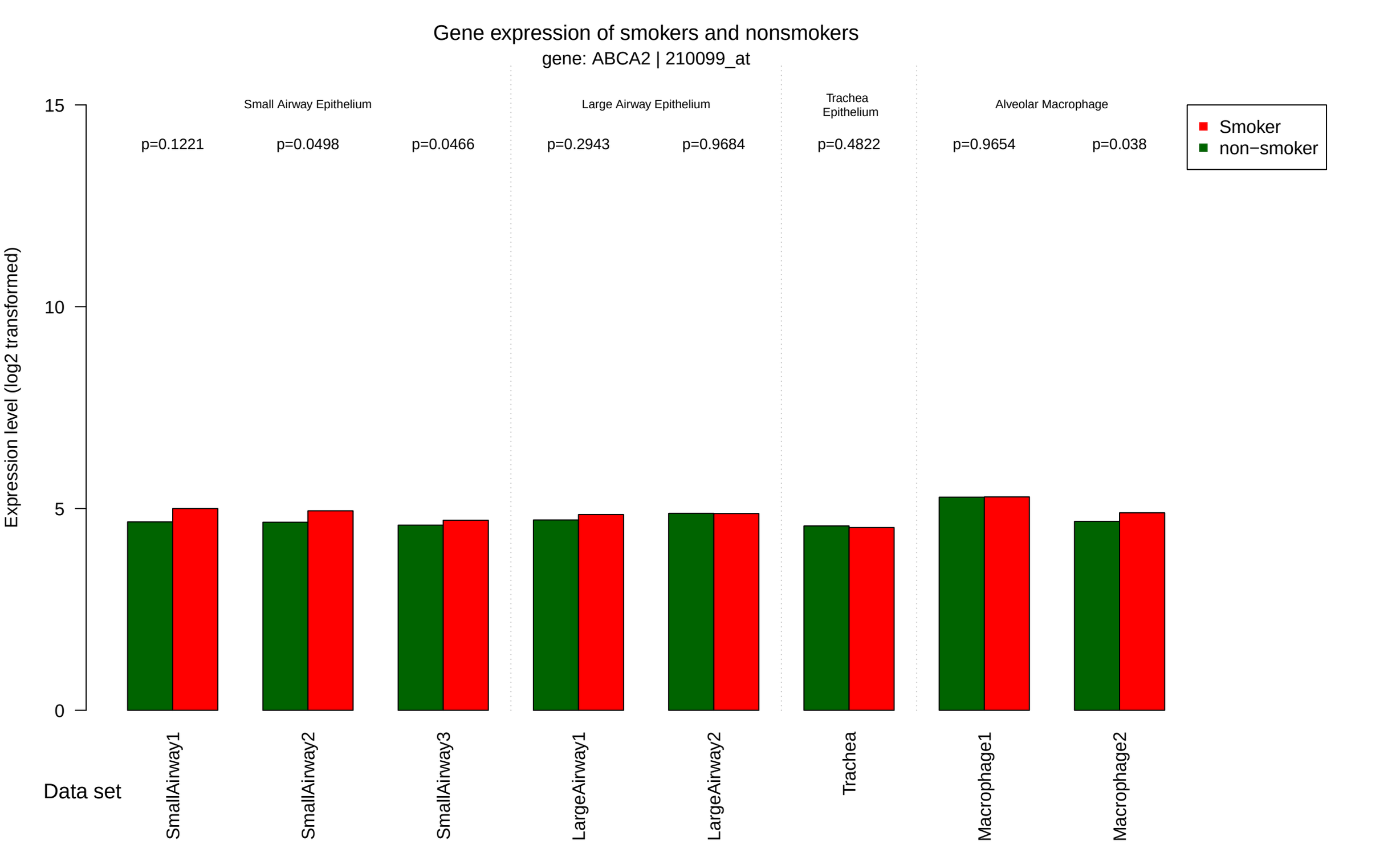
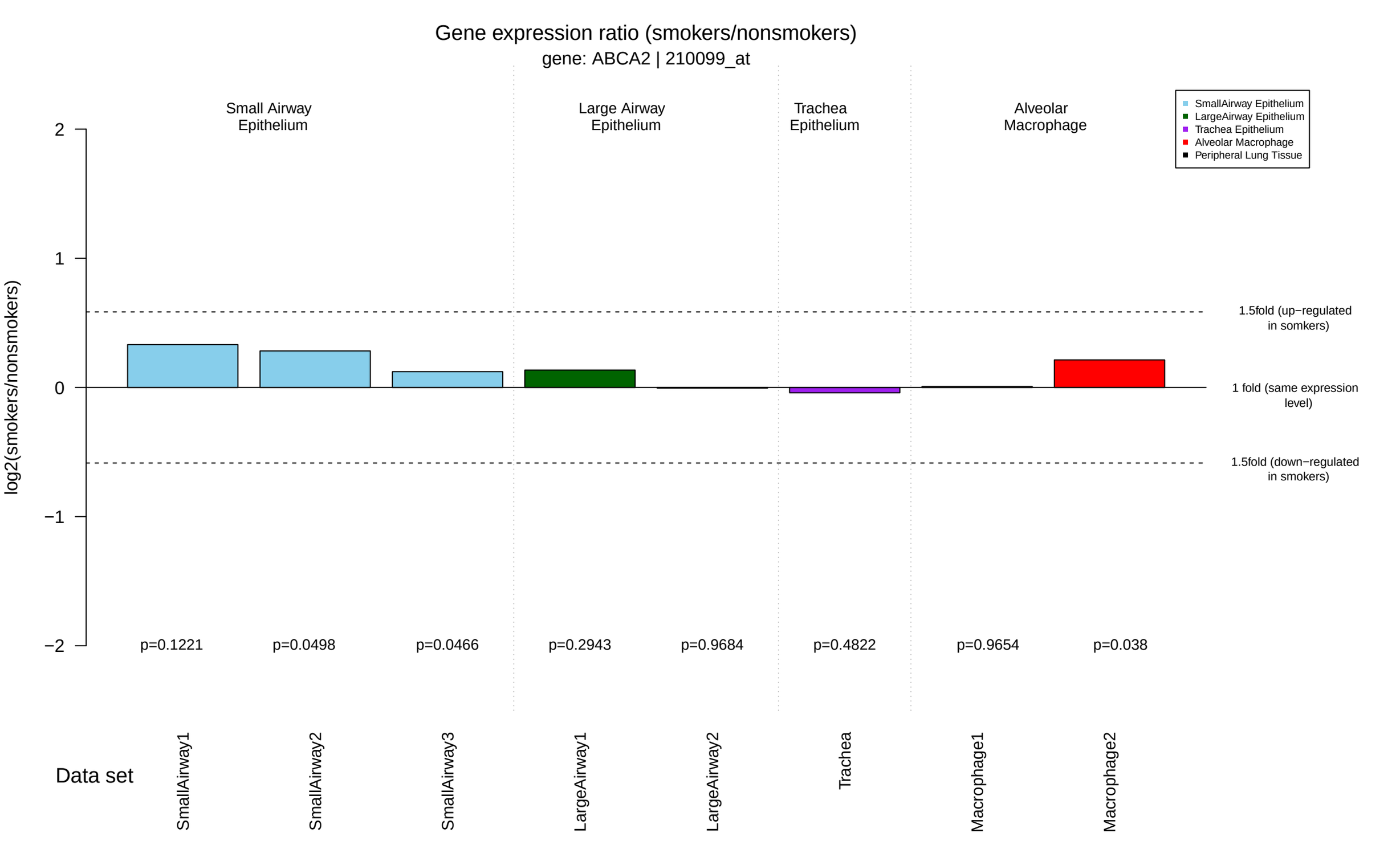
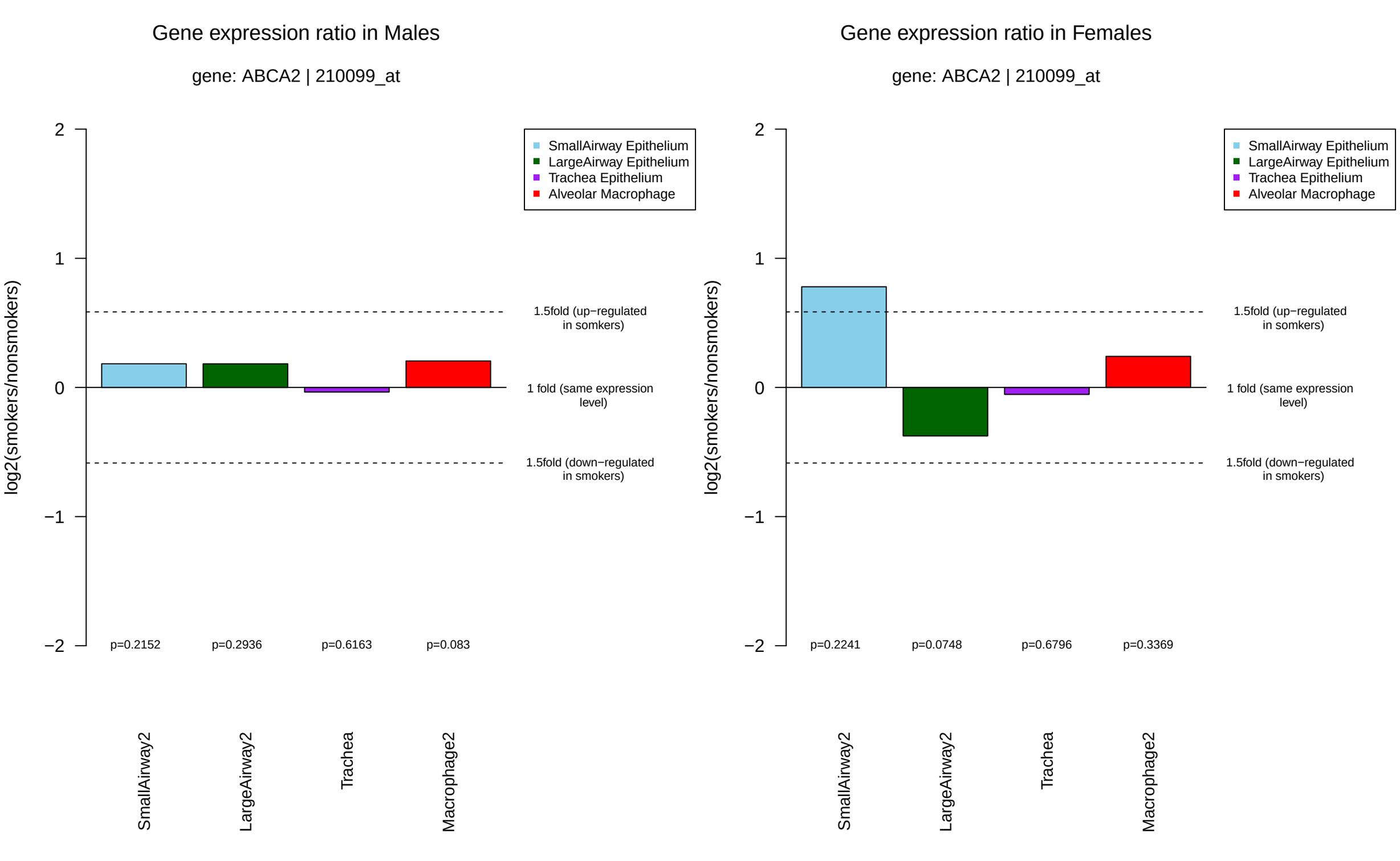
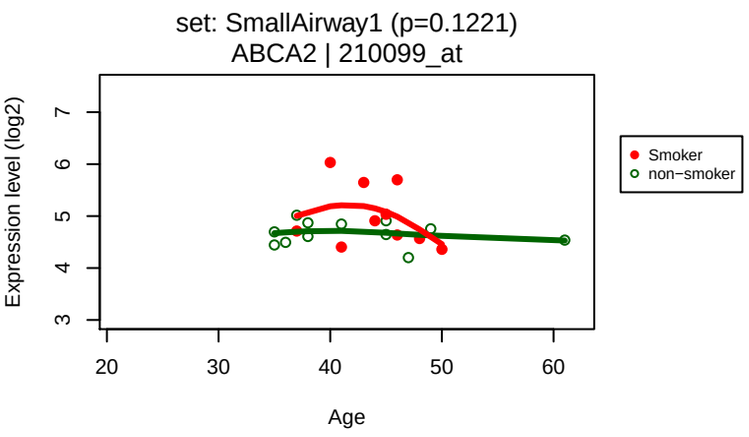
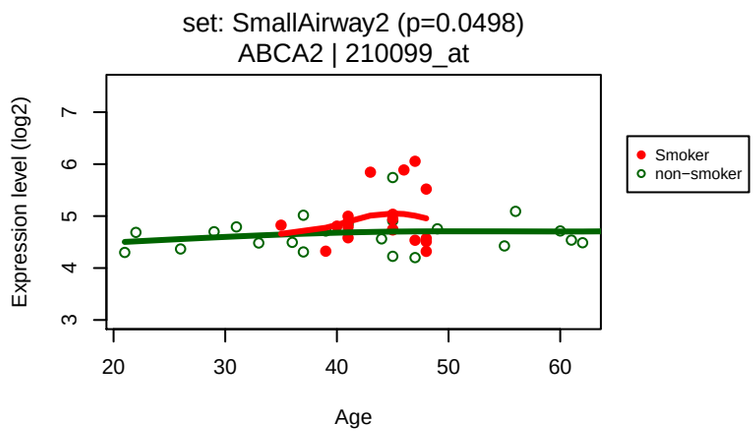
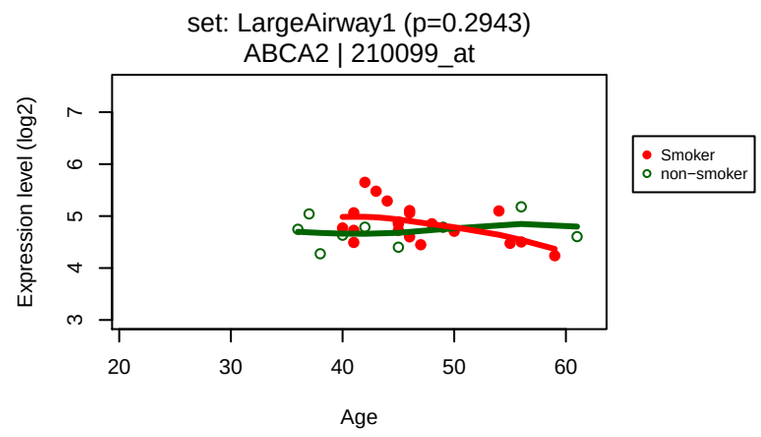

Gene expression of smokers and nonsmokers
gene: ABCA2 | 210099_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1221
p=0.0498
p=0.0466
p=0.2943
p=0.9684
p=0.4822
p=0.9654
p=0.038
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

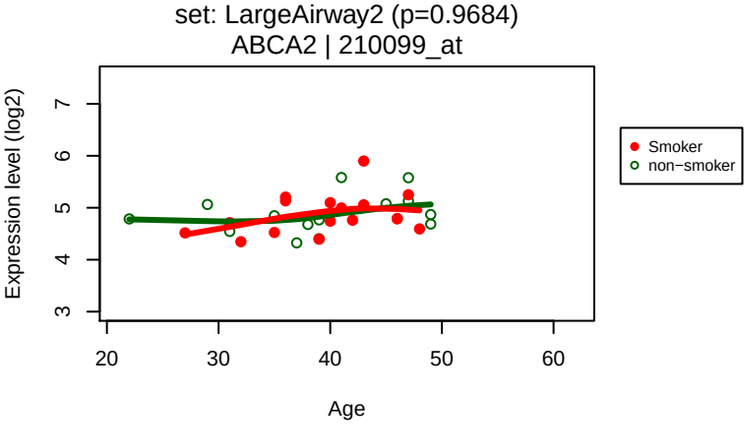

Gene expression ratio (smokers/nonsmokers)
gene: ABCA2 | 210099_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1221
p=0.0498
p=0.0466
p=0.2943
p=0.9684
p=0.4822
p=0.9654
p=0.038
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

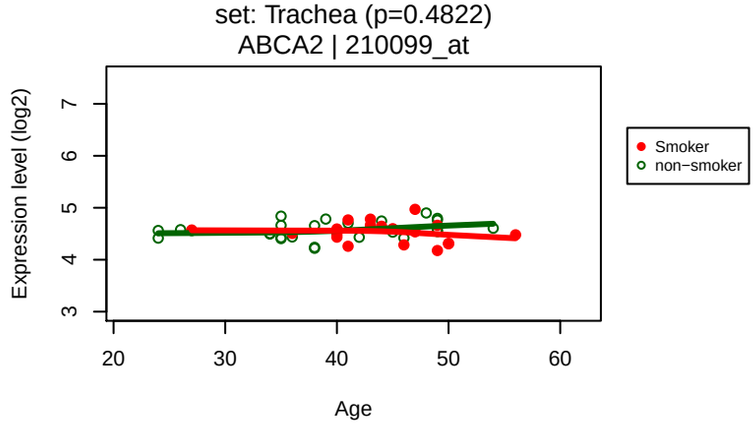

Gene expression ratio in Males
gene: ABCA2 | 210099_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2152
p=0.2936
p=0.6163
p=0.083
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ABCA2 | 210099_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2241
p=0.0748
p=0.6796
p=0.3369
SmallAirway2
LargeAirway2
Trachea
Macrophage2

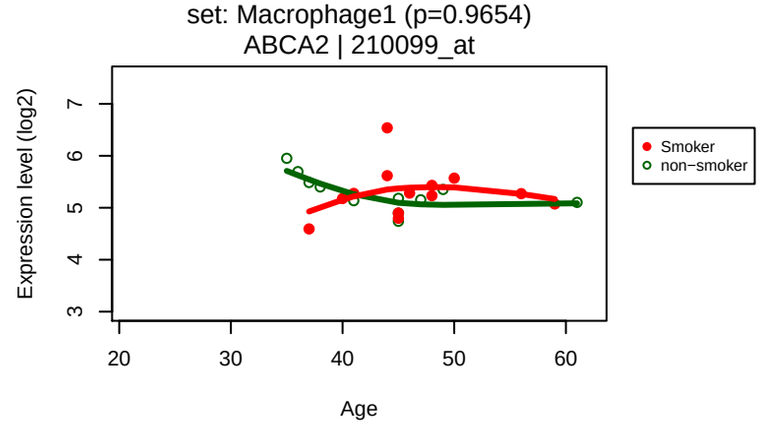

set: SmallAirway1 (p=0.1221)
ABCA2 | 210099_at
Expression level (log2)
Age
Smoker
non−smoker

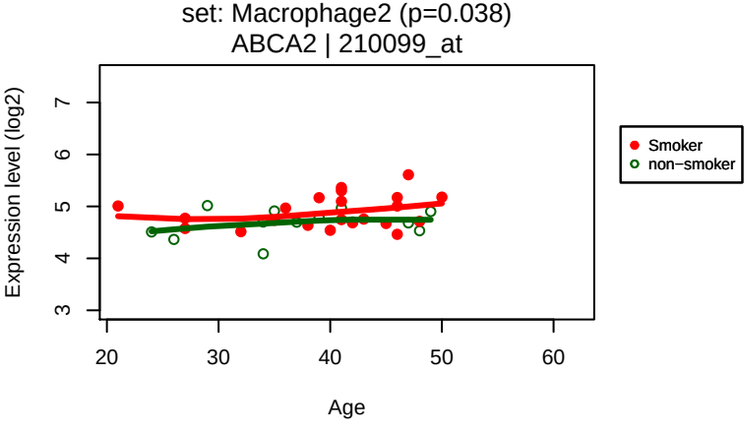

set: SmallAirway2 (p=0.0498)
ABCA2 | 210099_at
Expression level (log2)
Age
Smoker
non−smoker

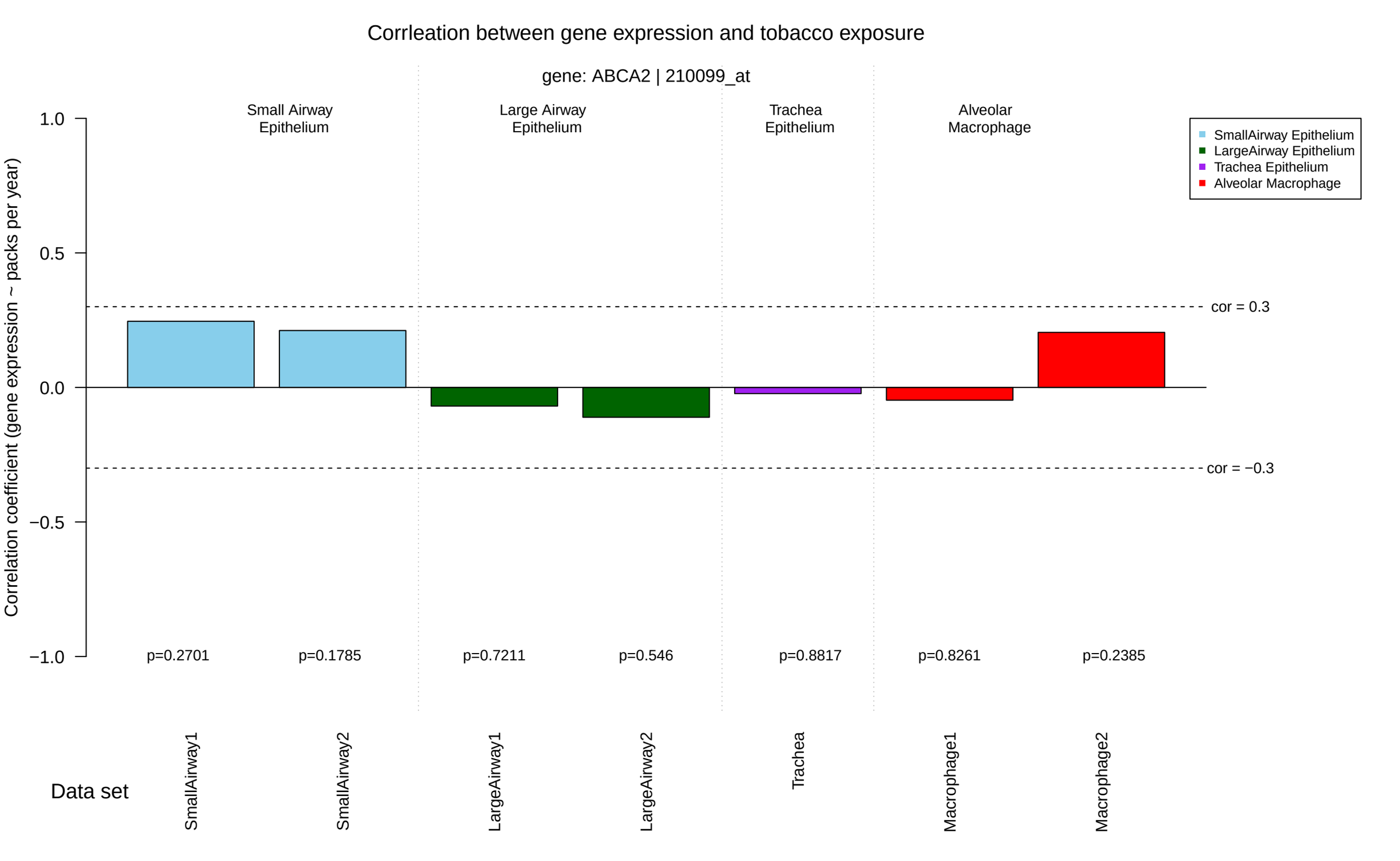

set: LargeAirway1 (p=0.2943)
ABCA2 | 210099_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9684)
ABCA2 | 210099_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4822)
ABCA2 | 210099_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9654)
ABCA2 | 210099_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.038)
ABCA2 | 210099_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ABCA2 | 210099_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2701
p=0.1785
p=0.7211
p=0.546
p=0.8817
p=0.8261
p=0.2385
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2